

Saakshi (साक्षी, "witness")

Solution synopsis. A resilient, trustworthy online exam (CBT) system for Challenge 6.

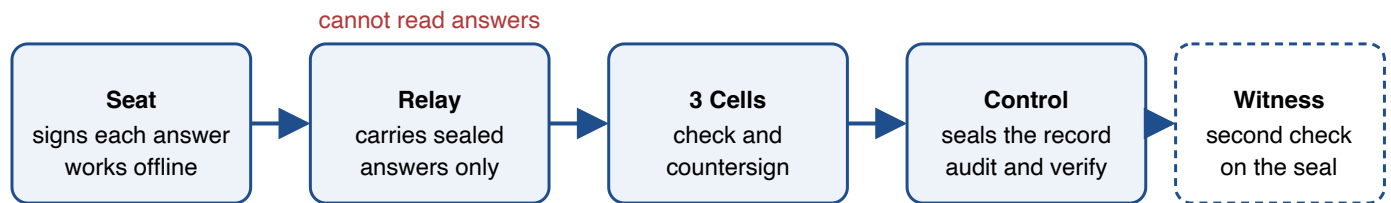
THE PROBLEM

Online exams can fail, and when they do, nobody can prove what really happened.

OUR SOLUTION

Sign every answer at the seat, keep going through failures, and make any tampering easy to find and fix.

How an answer travels



Three promises

We cannot promise zero failures. We make every failure:



Recoverable

Lose a cell, a relay or a link and nothing is lost.



Contained

One broken part stays one broken part. Other centres carry on.



Provable

Any edit after the fact is located, and the true answer is shown.

Numbers from our own tests

0

answers lost in 16,000 sent

20/20

chaos drills passed

1,000

of 1,000 tampered records located

~1 s

to recover (median 0.96 to 1.23 s)

19,803

of 19,803 candidates unlocked

227,882

entries stored, 0 rejected

44 ms

median answer delivery (p99 168 ms)

2.3%

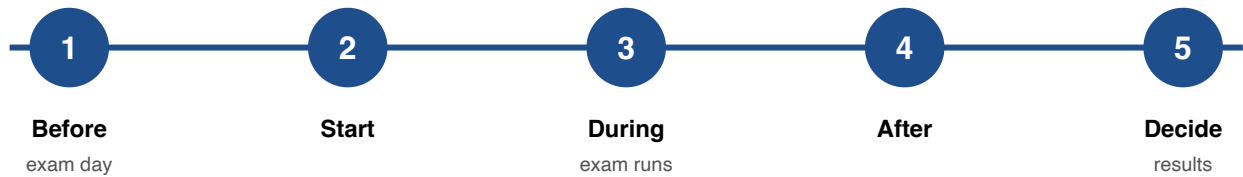
median CPU used by the seat app

Measured on one Apple M5 Mac, 3 real cells, 99 simulated centres, synthetic data. Not a real wide-area network.

Tested on macOS and Windows. Code and evidence: <https://github.com/anirudh12032008/saakshi>

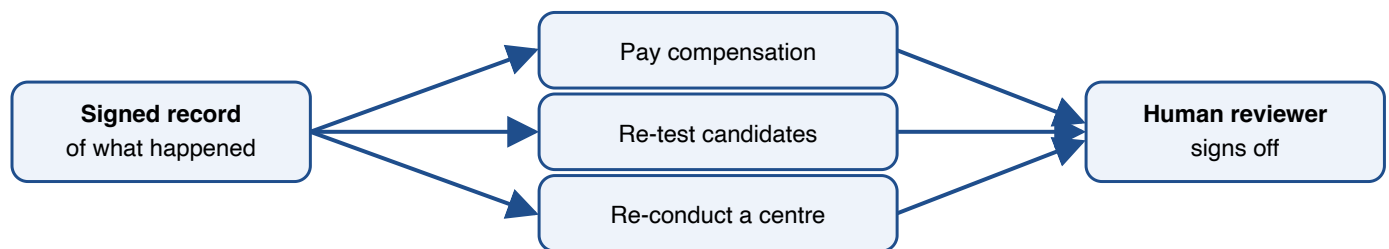
The five demo acts

Each act runs real software, not mocks, and leaves an evidence file.



Act	What we show	Result
1 Before Gate check	A seat running AnyDesk and an overlay tool is blocked by name. Close them and it turns green. A face flag goes to a human reviewer.	2 tools named and blocked. 1 flag cleared by a person.
2 Start Key release	2 of 3 key holders release the paper keys. 6 of 7 centres unlock. Centre 42 has a cut link, so it stays locked until a phoned code unlocks it. A forged key is rejected.	308 candidates, 7 centres, 6,586 entries saved.
3 During Chaos	We cut a centre's link, wipe a data centre cell, and force-quit a seat. Candidates keep answering. The wiped cell rebuilds. The seat resumes on another PC.	Answers lost: 0. Recovery 1.3 to 1.9 s.
4 After Tamper audit	A rogue insider edits an answer in the database. The audit finds it. The verify page shows "record says C, the seat committed B" and restores B.	1,000 of 1,000 edits located.
5 Decide Analytics	Analytics on the cells' own data flag likely cheating and choose: pay compensation, re-test, or re-conduct a centre. A human always signs off. No candidate is penalised automatically.	53 flags, no honest candidate flagged (synthetic data).

What happens to a failed exam?



One honest limit: a seat signature proves an answer was not changed after the device made it. It does not prove the candidate chose it. All analytics numbers use synthetic data.

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